

Lab 1 - Introduction to AWS Identity and Access Management (IAM)

Title

Introduction to AWS Identity and Access Management (IAM)

Objective

The objective of this lab is to understand how AWS Identity and Access Management (IAM) controls authentication and authorization in AWS. The lab focuses on exploring IAM users and groups, analyzing attached policies, assigning users to appropriate groups based on organizational roles, and validating permissions by testing service access.

Prerequisites

- Basic understanding of cloud computing concepts
- AWS Academy Lab access
- Web browser with internet connectivity

Tools Used

- AWS Management Console
- AWS Identity and Access Management (IAM)
- Amazon EC2
- Amazon S3

Tasks Performed

Task 1: Explore IAM Users and Groups

- Reviewed pre-created IAM users: user-1, user-2, user-3
- Explored IAM groups: EC2-Admin, EC2-Support, S3-Support
- Inspected managed and inline policies attached to groups

Screenshot:

IAM users (4) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Acc
☐	awsstudent	/	Access denied	-	Access denied	Access denied	-	Access denied
☐	user-1	/spl66/	0	-	-	9 minutes	-	Act
☐	user-2	/spl66/	0	-	-	9 minutes	-	Act
☐	user-3	/spl66/	0	-	-	9 minutes	-	Act

Task 2: Add Users to Groups

- Added user-1 to the S3-Support group
- Added user-2 to the EC2-Support group
- Added user-3 to the EC2-Admin group

Screenshot:

S3-Support [Info](#)

1 user added to this group.

Summary

User group name S3-Support	Creation time August 06, 2026, 13:17 (UTC+05:30)	ARN arn:aws:iam:199453238125:group/spl66/S3-Support
-------------------------------	---	--

Users (1) | Permissions | Access Advisor

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

☐	User name	Groups	Last activity	Creation time
☐	user-1	1	None	11 minutes ago

The screenshot shows the AWS IAM console interface. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access Management, Access reports, and various IAM tools. The main content area displays the 'EC2-Support' user group details. A green banner at the top indicates '1 user added to this group.' The 'Summary' section shows the user group name 'EC2-Support', creation time 'August 06, 2026, 13:17 (UTC+05:30)', and ARN 'arn:aws:iam::199453238125:group/spl66/EC2-Support'. The 'Users (1)' tab is active, showing a table with one user: 'user-2'. The table columns are User name, Groups, Last activity, and Creation time.

User name	Groups	Last activity	Creation time
user-2	1	None	12 minutes ago

The screenshot shows the AWS IAM console interface for the 'EC2-Admin' user group. Similar to the first screenshot, it shows the 'Summary' section with the user group name 'EC2-Admin', creation time 'August 06, 2026, 13:17 (UTC+05:30)', and ARN 'arn:aws:iam::199453238125:group/spl66/EC2-Admin'. The 'Users (1)' tab is active, showing a table with one user: 'user-3'. The table columns are User name, Groups, Last activity, and Creation time.

User name	Groups	Last activity	Creation time
user-3	1	None	12 minutes ago

Task 3: Test IAM User Permissions

- Logged in using IAM sign-in URL
- Verified S3 access for user-1
- Verified EC2 read-only access for user-2
- Verified EC2 administrative access for user-3

Screenshot:

Failed to stop the instance i-0dbcac18181d77574

You are not authorized to perform this operation. User: arn:aws:iam::199453238125:user/spl66/user-2 is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:us-east-1:199453238125:instance/i-0dbcac18181d77574 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: K6FTCFNB8MR45D8TeBwMFCT_RccpTftSmZQUC-b5fs-LT-C1dTelLJJJzolsTWjWA4khuTlrlW9ndPygy1gGr_ArPYBIBKqEv-58RjgyaAlOPCO8hd83jBh9GvHfOQcQyddp-9nv8j52A12v-1KXr-Cda4l2lqKFVxo9GxhPdtbzhBoNshfCpeweljgm2gKnuul95gvg4qfzPogLgRqOSHgslv7czp6s1j3X4YAb8l5qDSLcv_rL_7Yj6ad9WUqUJkKw-kU22FPSE7A307jQI6ZDYjTHOktrD-szJuyALgp9Iodp1n_OGSJwJRdwp3w3b5bwhfg4xpjggBnlkt45Hic6K6pmuWsy__JN-jWqj2vn2CB90AM6e24tftbuMXvNlPbFovzJbT2_0JuRXdqFBGK_KHweJ1DMLUrcFDH9o9PX4GIMjFCnJDovrowXpaYphMokTzld1d1ZRNfY_dRZJIPBRdHjtUE4ipZKxJBhqlBTelEzO77V_u7TlHuxqlgwjOs3rYkBgMwidTAQoNNajJmgyTx-wlbhRG8ZL_KOpnFGCNkbcMk2g66PDkn-OkRWjYjCALK2aEW0HdD5ggvVHXZMSrp2V5Fkel40L-JMDQjO5zDUOaVoV-2SSsowdzYaaL-EqJEQCBvjGcTXlxZ67fGatfWlosRjE90n-fCBtlqWzSQ3G29IYsU1yJUKsXslsCKHb4VXdy2PDQZciqYtnnJM6eORQD2khfMynYDNDwc4LMwlu7VGRD8578_pkM2ouWdSpClt-c5aJawY1V-aR9KqjBDZ_81jpPpWk8DEoV3sJem07y6vZFSQzpvFu7bKfXshsvCldlpA7INQjrmKk70IEcsSdU_xp-lj3-3ldnyBUxkMIL7xl-hwMTgrudsmIq9zprJmoaRTIYU3p_hUyVbQxlmHwaH0yGpL4FGbCKCmEMH0QvwZklg8telPvQLDKFH29EO5I9GDDUdhlogmWSxruox-Vfd53uWXPJMJUVikwp6vaNlf3388Rp5at4qA8psbEZIXh7gUwMg58MyQuOn7A5VA

Instances (1/2) Info

i-0dbcac18181d77574 (LabHost)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID i-0dbcac18181d77574	Public IPv4 address 54.162.65.60 open address	Private IPv4 addresses 10.1.11.157
IPv6 address -	Instance state Running	Public DNS ec2-54-162-65-60.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-1-11-157.ec2.internal	Private IP DNS name (IPv4 only) ip-10-1-11-157.ec2.internal	

Successfully initiated stopping of i-0dbcac18181d77574

Instances (1/2) Info

Saved filter sets: Choose filter set | Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Bastion Host	i-0a7962269f4164c74	Running	t2.micro	2/2 checks pass	us-east-1a	ec2-54-208-120-88.co...
LabHost	i-0dbcac18181d77574	Stopping	t2.micro	-	us-east-1a	ec2-54-162-65-60.com...

i-0dbcac18181d77574 (LabHost)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID i-0dbcac18181d77574	Public IPv4 address 54.162.65.60 open address	Private IPv4 addresses 10.1.11.157
IPv6 address -	Instance state Stopping	Public DNS ec2-54-162-65-60.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-1-11-157.ec2.internal	Private IP DNS name (IPv4 only) ip-10-1-11-157.ec2.internal	

Workflow

1. Accessed IAM console and reviewed users and groups.
2. Inspected policy permissions attached to groups.
3. Assigned users to groups based on their roles.
4. Logged in as each IAM user using the sign-in URL.
5. Validated permissions by accessing AWS services.

Learning Outcomes

- Understood the role of IAM in AWS security.
- Learned how IAM users, groups, and policies interact.
- Gained practical experience implementing role-based access control.
- Verified permission enforcement through real-time service testing.

Conclusion

This lab provided hands-on experience with AWS IAM by demonstrating how organizations manage secure access to cloud resources. Assigning users to groups with predefined policies simplified permission management and ensured role-based access control across AWS services.

Author

Name: Junaid Sardar S (212224100028) **Course:** Introduction to Cloud Computing